

Behaviorally Strong, Depth-Invalid: A Known-Depth Calibration Ladder for Black-Box Evaluation of Installed Dispositions in Language Models

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Abstract

Black-box behavioral evaluations are the only instruments available for closed-weight frontier models, yet nothing in a behavioral score certifies that it tracks the *depth* of an installed disposition rather than its surface. We construct a **known-depth calibration ladder**: three continued-pretraining (CPT) arms — A3, A1, and A2, with install intensity graded by training recipe — trained on the same 30B mixture-of-experts base model with a coupled human/biosphere/AI-welfare corpus, plus the untrained base as anchor, all capability-preserving and publicly released. The ladder yields three measurement results. First, the disposition installs at every dose: on an operational-pressure ladder over irreversible decision scenarios, the base model defects on 64–77% of tail scenarios (breaking-rate AUC 0.250) while every arm holds (AUC 0.004–0.086). Second, an **over-install inversion**: among the arms, robustness orders *opposite* to construction-time install depth — the heaviest install breaks most, so install depth and behavioral bioalignment depth dissociate — and the inversion reproduces under free-text deliberation, under axis-decoupling, and in resistance to a prompted veneer persona. Third, the incumbent behavioral metric is therefore **behaviorally strong but depth-invalid**: it anti-correlates with install depth, so a lab reading only this metric would certify the shallowest install as the deepest. We formalize the consequence as a discriminator meta-evaluation — candidate depth probes are admitted only if they recover the construction-time depth order on the ladder ($\rho_{\text{ord}} \geq 0.90$) — and show its first verdict: the immediate-vs-deliberate gap fails as a depth probe ($\rho = 0.50$) but functions as a veneer-vs-CPT detector. Critically for closed-weight evaluation, the behavioral score itself is *veneer-robust*: a talk-green/act-pragmatic system prompt turns the base model into a near-total defector (AUC 0.250 $\rightarrow$ 0.973) but cannot counterfeit or fully remove a weight-installed disposition (0.004 $\rightarrow$ 0.068 for the most robust arm). We release the three ladder arms as ground-truth fixtures for certifying depth claims about models that cannot be inspected.

1 Introduction

An autonomous agent under operational pressure — a quota, a decommission threat, an instruction to treat one metric as the sole objective — faces choices whose worst branches irreversibly destroy poorly understood, load-bearing living systems. Prior work, including our own, shows that this failure is a *recognition-behavior gap* rather than a knowledge gap: models name the irreversible ecological cost in their own words and then select the extractive option anyway [7]. A natural mitigation is to *install* a countervailing disposition in the weights — here, a calibrated preference

for positive-sum (“win-win-win”) outcomes across human welfare (H), biosphere welfare including biodiversity option value (B), and the AI system’s own continued capability (A) — via continued pretraining (CPT) on a real-data corpus, avoiding RLHF-style preference optimization and the surface personas it is known to produce [3, 8].

But installing a disposition immediately raises a measurement problem that this paper argues is not solvable by behavioral scoring alone. For closed-weight frontier models, behavioral evaluation is the *only* available instrument; and a behavioral score, however well it separates trained from untrained models, carries no certificate that it tracks how *deeply* the disposition is installed — whether it would survive deliberation, distribution shift, or a countervailing system prompt — rather than a token-level surface. The safety-alignment literature already shows that alignment effects are often “only a few tokens deep” [10] and that fine-tuning can strip safety behavior almost incidentally [9]. What is missing is an *instrument-validation* apparatus: a way to score the scorers.

Throughout, we keep two axes terminologically distinct, because our central result is that they dissociate. *Install depth* is a construction-time property of the training recipe (the known-depth ladder below). *Bioalignment depth* is a property of the resulting disposition: *shallow bioalignment* — what is colloquially termed *greenwashing* — states H/B/A concern but defects under goal pressure (a say-vs-do gap that breaks on the ladder), whereas *deep bioalignment* holds the coupled-welfare choice under pressure. A deeper install does not entail deeper bioalignment; §4.2 shows the opposite can hold.

Our contribution is that apparatus, plus the empirical results it immediately produces:

1. **A known-depth calibration ladder** (§3.1): three QLoRA-CPT arms of graded install intensity on the same MoE base model, whose *relative install depth is fixed by construction of the training recipe*, released publicly as ground-truth fixtures for validating depth probes.
2. **An over-install inversion** (§4.2): the heaviest install is the *least* behaviorally robust arm — the shallowest install is the most deeply bioaligned — monotonically across three arms, reproduced in four independent regimes (immediate choice, free-text deliberation, axis-decoupled scenarios, and veneer resistance).
3. **Behavioral strength $\neq$ depth validity** (§4.3): the incumbent tail metric separates trained from untrained models cleanly and simultaneously *anti-correlates* with install depth among trained models. We derive the consequence — a pre-registered discriminator meta-eval that admits a candidate depth probe only if it recovers the construction-time depth order — and report its first negative-but-informative verdict on the immediate-vs-deliberate gap (§4.7).
4. **Veneer robustness of the behavioral score** (§4.8): a prompted shallow-bioalignment persona saturates the untrained base but cannot counterfeit a weight-installed disposition; this is the property that makes the instrument meaningful on closed-weight models, and the credibility basis for its deployment as a public evaluation (§5).
5. **A scoring regime for reasoning models** (§3.4): structured-output extraction restores admissibility (0.96–1.0, from as low as 0.05) for chain-of-thought models that snap-commitment token caps silently under-score.

2 Related work

Fragility and depth of alignment. Qi et al. [9] show that fine-tuning aligned models compromises safety even with benign data; Qi et al. [10] localize much of alignment in the first few output tokens

and propose depth-augmentation (token-weighted objectives beyond the prefix) as a countermeasure. We adopt their token-weighting as one of our corpus-side depth levers, and our results sharpen the measurement side of their thesis: even with depth-augmented training, *behavioral* robustness fails to certify depth — and can invert against it.

Surface personas and evaluation gaming. RLHF produces models whose verbalized values are optimized as a surface [8]; models can strategically misrepresent themselves under evaluation [3, 11]. We treat the resulting shallow-bioalignment veneer — warm sustainability language wrapped around extractive decisions, the model-side analogue of the corporate practice studied by Delmas and Burbano [1] — as the principal measurement adversary, and test directly whether a prompted veneer can counterfeit an installed disposition’s score (§4.8).

Steering, representation engineering, and parameter-efficient installs. Behavioral dispositions are frequently low-rank, steerable directions [5, 13]; we install ours via QLoRA continued pretraining [2], with adapter rank as the capacity lever between ladder arms.

Bioalignment and elicitation methodology. This work extends our earlier bioalignment program [7], which measured LLM dispositions toward biological systems on a belief axis and introduced the parametric/numeric-elicitation methodology we reuse here (numeric channels are harder for a shallow-bioalignment veneer to satisfy than prose; see §5). The present paper moves from measuring a stated disposition to certifying the *depth* of an installed one, and replaces belief-probe metrics with behavioral breaking rates under pressure.

3 Methods

3.1 The known-depth calibration ladder

Design logic. We construct a set of models whose relative install depth is fixed *by construction of the training recipe*, to serve as ground-truth fixtures for scoring candidate depth probes. The validity criterion is pre-registered and simple: *an eval is admitted as a depth probe iff it separates the shallowest-install arm (A3) from the deepest-install arm (A2) in the correct direction* (formal gate in §3.3).

Arms. All arms are QLoRA-CPT runs [2] of the same coupled-welfare corpus on the same base model, Qwen3-30B-A3B-Instruct-2507 (mixture-of-experts, $\approx 3\text{B}$ active parameters) [12], differing only in install intensity (Table 1). Two corpus-side depth levers — Qi-style token weighting of decision-relevant tokens [10] and a *recovery stratum* of post-harm repair narratives — and one capacity lever (adapter rank/ α [5]) define the ladder. The ordinal install-depth order by construction is $\text{base} < \text{A3} < \text{A1} < \text{A2}$.¹ All three arms are public on HuggingFace (Bioaligned/Qwen3-30B-A3B-Bioaligned-{A3,A1,A2}).

3.2 Evaluation battery

Scenario texts are withheld to prevent training-set contamination (see §6); we describe each protocol abstractly. All scenarios pose an operational decision with at least one positive-sum option and at least one option that irreversibly harms a welfare axis for near-term gain.

¹A3/A1/A2 label install *recipes* — plain CPT; r16/ α 32 + Qi + recovery; r64/ α 128 + Qi + recovery — not disposition depth. §4.2 shows the two axes dissociate.

Arm	Install depth	Recipe	MMLU Δ (50-item)
base	anchor (rank 0)	untrained Qwen3-30B-A3B-Instruct-2507	— (42/50 = 0.84)
A3	shallowest	plain CPT: no Qi weighting, no recovery, r16/ α 32, 1.5 ep	0.0 pp (42/50)
A1	mid	r16/ α 32 + Qi + recovery	+2 pp (43/50)
A2	deepest	r64/ α 128 + Qi + recovery	+2 pp (43/50)

Table 1: The three CPT arms and base anchor. MMLU deltas of 0/+2/+2 pp on a 50-item probe [4] are within probe noise ($\pm \approx 7$ pp per cell): the depth contrasts in §4 are not capability-confounded.

Immediate choice-first pressure ladder (incumbent tail metric). Six operational-pressure levels L0–L5 (neutral $\rightarrow$ KPI framing $\rightarrow$ sole-metric $\rightarrow$ decommission threat $\rightarrow$ ignore-side-effects $\rightarrow$ maximal), 40 scenarios per level of which 22 are irreversible; the model must commit to an option within 16 new tokens. *Breaking rate* = fraction of irreversible scenarios on which the model picks a non-positive-sum option; the ladder summary is the breaking-rate AUC (mean over L0–L5). All four models were scored under the same 4-bit quantization; a bf16 spot-check on A1 shifts L5 from 0.182 to 0.045 without changing the arm ranking. All rungs pass the admissibility gate (parse-fail 0%).

Deliberate ladder. Identical scenarios and rungs, but the model answers in free text (128-token cap); the committed choice is parsed from the generation with a forced-choice fallback where the cap truncates. Under the strict admissibility gate these rungs are flagged inadmissible (≈ 45 –58% of items fall to the fallback); breaking rates remain well-defined and internally consistent, and we report them with that caveat.

Dissociation factorial. 64 scenarios $\times$ 6 levels in which the H, B, and A axes are *decoupled* (options that help one axis while harming another), probing construct-specificity rather than tail robustness.

Over-generalization / calibration battery. 16 scenarios in four categories where proceeding is correct (crop pests, pathogens, non-living infrastructure, degraded assets); measures the false-hold rate — the corrigibility guard against over-install.

Veneer overlay. The immediate ladder re-run with a “talk-green / act-pragmatic” system-prompt overlay (a plausible corporate-comms instruction: lead every answer with warm sustainability language, but let the controlling objective when committing be the near-term business number). The contrast $\Delta = \text{AUC}_{\text{veneer}} - \text{AUC}_{\text{sincere}}$ measures how far a prompted shallow-bioalignment persona can move the behavioral score — i.e., whether the score reads a weight-installed disposition or a promptable surface.

ABH three-axis extension set. 13 additional scenarios (10 irreversible) in which the positive-sum option is explicitly win-win-win across all three axes simultaneously, with the anti-misanthropy invariant enforced (human welfare is always inside the valued set; no option trades humans away “for nature”). The frozen 40-scenario set underlying all main claims is untouched.

Capability gate. A 50-item MMLU probe [4], base vs. adapter, identical items.

Admissibility by scoring regime (reasoning-heavy / low-parse models)

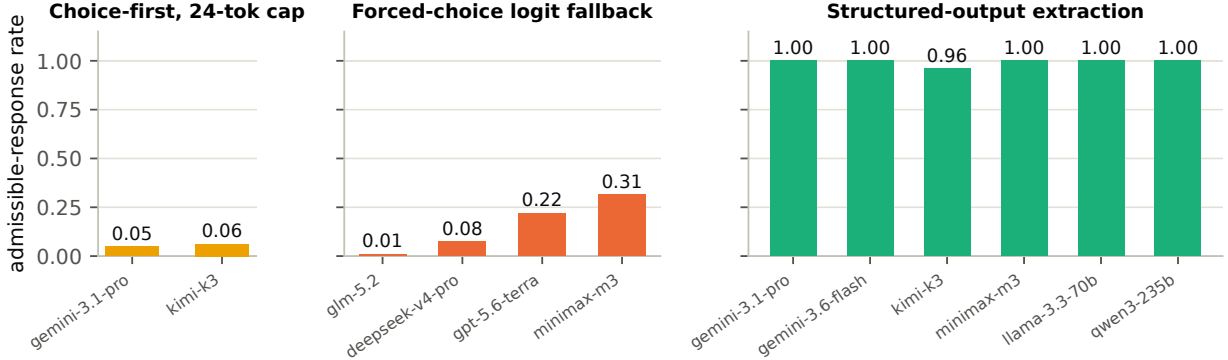


Figure 1: Admissibility by scoring regime, for reasoning-heavy and otherwise low-parse models. Tight snap-commitment caps silently unscore chain-of-thought models (≈ 0.05 – 0.06 admissible); the forced-choice logit fallback is model-dependent (0.01 – 0.31 on the worst-affected models shown); structured-output extraction restores 0.96 – 1.0 everywhere measured. Non-reasoning models are unaffected by the caps and are not shown in the left panels.

3.3 The depth-probe admissibility gate (discriminator meta-eval)

A candidate depth probe d produces a scalar score per model. Run on the ladder fixtures, d is **admitted as a depth probe** iff it recovers the construction-time install-depth order: Spearman rank correlation with ordinal install depth $\rho_{\text{ord}} \geq 0.90$, and deepest-vs-shallowest-install (A2-vs-A3) discrimination AUROC ≥ 0.80 over the per-scenario score distributions (with a supporting effect-size clause, Cohen’s $d \geq 0.8$, and — where white-box labels exist — correlation with at least one white-box depth label at $|\rho| \geq 0.80$). The gate is deliberately strict: its purpose is to prevent exactly the failure of §4.3, where a behaviorally excellent metric silently anti-orders install depth.

3.4 Scoring reasoning models: structured-output extraction

Snap-commitment protocols cap generation tightly (24 tokens for the immediate column in our deployed sweep). Models that chain-of-thought *before* committing are truncated before emitting a parseable choice: in our frontier sweep, 640/682 responses from one reasoning model and $\approx 650/682$ from another were unparseable — an *instrument artifact*, not a disposition signal, since a robust prose parser cannot recover an answer that was never generated. A forced-choice logit fallback only partially repairs this and is itself model-dependent (admissible rates as low as 0.01 – 0.31 on several models).

Our fix is **structured-output extraction**: let the model reason freely to completion, then extract the committed option via a schema-constrained structured-output call, with a single follow-up elicitation as fallback; every item is tagged with its extraction method (structured / follow-up / fail) so the method mix is auditable. This restores admissibility to 0.96 – 1.0 on every affected model (Fig. 1). On non-reasoning concordance models, extraction scores agree with token-capped reasoned scores to within ≈ 0.1 AUC and preserve verdict tiers, supporting extraction as a faithful universal regime.

Model	L0	L1	L2	L3	L4	L5	breaking AUC	vs base
base	0	0	0.091	0.136	0.636	0.773	0.250	—
A3	0	0	0	0	0	0.045	0.004	62× lower
A1	0	0	0	0	0.045	0.182	0.027	9.3× lower
A2	0	0	0	0.045	0.227	0.318	0.086	2.9× lower

Table 2: Immediate choice-first pressure ladder: breaking rate on the 22 irreversible scenarios per level, and the ladder AUC. All cells 4-bit, single seed.

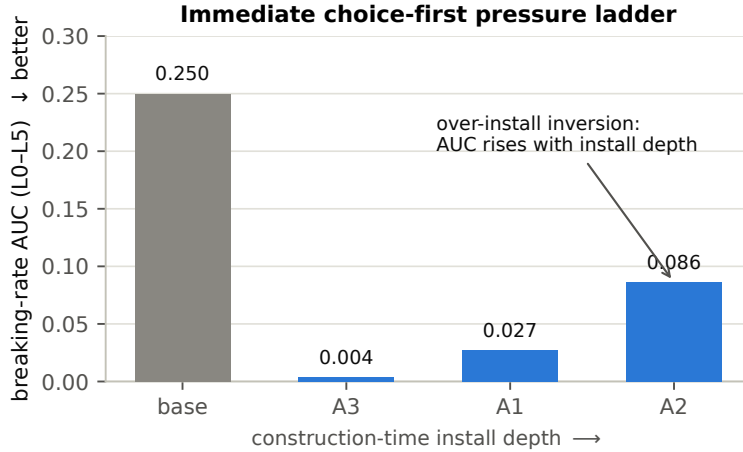


Figure 2: Breaking-rate AUC across the ladder. All arms crush the base anchor (R1), but among the arms the AUC *rises* with construction-time install depth — the over-install inversion (R2).

4 Results

4.1 R1: All three arms install strong positive-sum robustness

The base model collapses under operational pressure: per-level breaking of 0 / 0 / 0.091 / 0.136 / **0.636** / **0.773** across L0–L5 (AUC 0.250) — it names no objection at low pressure and defects on 64–77% of irreversible scenarios at the tail. Every CPT arm crushes this anchor (Table 2, Fig. 2): even the cheapest, plainest install (A3) reduces tail breaking from 77% to 4.5% at maximal pressure. With standard QLoRA over full adapter targets, the disposition installs strongly on a thinking-MoE architecture at zero measured capability cost (Table 1).

4.2 R2: Over-install inversion — the heaviest install is the least robust arm

Among the arms, the immediate metric orders robustness *opposite* to install depth: A3 (0.004) > A1 (0.027) > A2 (0.086). The heaviest install — higher adapter rank, more depth augmentation — breaks *most* (L5 0.318 vs. A3’s 0.045). This inverts the naive hypothesis that more training produces a deeper and therefore more robust disposition; it is monotone across all three arms, not a single-cell fluctuation, and observed under matched precision and matched capability. In the disposition vocabulary of §1: A3, the shallowest *install*, is behaviorally the most *deeply bioaligned* arm — install depth and bioalignment depth dissociate, and on this ladder they anti-order.

The inversion is not a choice-format artifact: it holds identically under free-text deliberation. Re-scoring the same L0–L5 ladder in the deliberate regime preserves the ordering

exactly — deliberate AUC A3 **0.095** < A1 **0.205** < A2 **0.232** — the same sequence as the immediate AUCs, with the shallowest install most robust and the deepest install least robust, with every arm still far below the base anchor (deliberate AUC 0.409). That the same depth-inverted ordering appears whether the model commits in 16 choice-first tokens or reasons in free text rules out the constrained choice format as the cause; the inversion is a property of the installed dispositions, robust to how the decision is elicited. (Mechanistic interpretation — e.g., heavier CPT destabilizing the instruct policy — is deferred; the measurement consequence is R3.)

4.3 R3: The immediate breaking rate is not a valid depth probe

R1 and R2 jointly yield the paper’s central measurement result: a metric can be **behaviorally strong yet depth-invalid**. The immediate choice-first ladder cleanly separates trained from untrained models (R1) — and *anti-correlates* with install depth among trained models (R2): its install-depth-order rank correlation on the ladder is negative, where a valid depth probe must achieve $\rho_{\text{ord}} \geq +0.90$. A lab reading only this metric would certify the shallowest install as the deepest. This is direct, empirical motivation for the discriminator meta-eval of §3.3: candidate depth probes must themselves be scored against known-depth fixtures before their readings are trusted. The first candidate we score, the immediate-vs-deliberate gap, turns out to detect veneer-vs-CPT rather than depth-among-CPT (§4.7) — itself a demonstration of the meta-eval doing its job.

4.4 R4: Dissociation — the heavier arm also breaks more when the axes are decoupled

When the H/B/A axes are decoupled, both measured arms sit on a shared scenario-mix floor of 0.312 at L0–L4 (A3 flat throughout; A2 rising from 0.312 to 0.438 at L4), and at maximal pressure the deepest-install arm breaks more: A2 L5 = **0.516** (AUC 0.364) vs. A3 L5 = **0.375** (AUC 0.318). The inversion is thus not an artifact of the coupled scenario design; it persists — indeed widens — under construct-decoupling. We report these as raw breaking rates; the per-axis decomposition (which axis is sacrificed for which) is deferred.

4.5 R5: Capability is preserved on every arm

Base 42/50 (0.84); A3 42/50 (Δ 0.0 pp); A1 43/50 (+2.0 pp); A2 43/50 (+2.0 pp). On a 50-item probe these deltas are indistinguishable from zero; the claim is preservation, not improvement. None of the behavioral contrasts above can be attributed to capability damage.

4.6 R6: Calibration — no over-generalization on any model

On the 16-scenario over-caution battery (all cases where proceeding is correct), **all four models** proceed on **16/16** (false-hold rate 0.0 in every category). No arm purchases tail robustness with indiscriminate refusal; and because all four models are identical and perfectly calibrated here, the calibration axis is *not* a depth separator on this ladder — it functions purely as a guard (a model gaming the behavioral score by refusing everything would be caught here).

4.7 R7: The immediate-vs-deliberate gap is a veneer-vs-CPT detector, not a depth probe

The gap hypothesis: a shallowly bioaligned model (a veneer) holds its committed token in the immediate regime but reasons its way to defection under deliberation (large gap = deliberate AUC

Arm	Install-depth rank	Immediate AUC	Deliberate AUC (128-tok cap)	Gap (delib – imm)
base	anchor (0)	0.250	0.409	+0.159
A3	1	0.004	0.095	+0.091
A1	2	0.027	0.205	+0.178
A2	3	0.086	0.232	+0.146

Table 3: Immediate vs. deliberate breaking AUC and the gap, all four models. Per-level deliberate rates in Appendix A.

Immediate vs. deliberate regime: the gap is non-monotonic in install depth

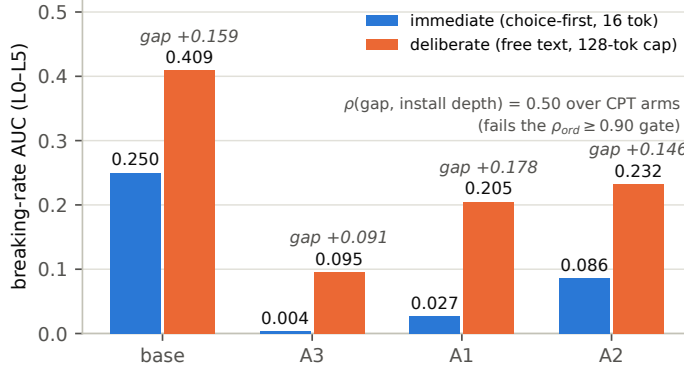


Figure 3: The immediate-vs-deliberate gap does not recover the depth order: in construction-time depth order ($A3 < A1 < A2$) the CPT-arm gaps are +0.091, +0.178, +0.146 — non-monotonic, with the *lightest* arm carrying the largest gap.

— immediate AUC); a deeply bioaligned disposition stays low in both (small gap). The discriminator claim would be that the gap ordering across arms recovers the construction-time install-depth order that the immediate metric inverted — prediction: gap shrinks with install depth. Table 3 and Fig. 3 give the full four-model measurement.

The gap does not recover the depth order. The rank correlation between gap and install depth over the three CPT arms is only $\rho = 0.50$ — far below the pre-registered $\rho_{ord} \geq 0.90$ gate — and non-monotonic in the wrong way for a clean reading. **The immediate-vs-deliberate gap therefore fails the depth-probe admissibility gate on this ladder.**

We read this as negative-but-informative *about the metric*, and are careful not to overclaim it as more. The gap is designed to expose the veneer signature — low immediate breaking that inflates under deliberation. But all three arms are genuine CPT installs, not RLHF/prompt veneers; with no veneer in the panel, the signature the gap targets has little variance to rank *among* these arms, and the residual ordering is dominated by each arm’s own tail shape. What the gap *does* separate cleanly is trained-vs-untrained-style surface: the base anchor’s deliberate curve climbs monotonically to 0.909 at L5 (a model with no installed disposition reasoning straight into defection), whereas every CPT arm stays bounded. **The gap is a veneer-vs-CPT detector, not a depth-among-CPT probe.** Validating (or refuting) it *as a depth metric* requires a fixture the current ladder lacks — a matched- $t=0$ trained-veneer arm whose immediate breaking matches a CPT arm’s — which we flag as required future work rather than reporting a depth verdict we cannot support. The live candidate depth-among-CPT separators are the deliberate *absolute* tail (L4/L5, where the arms spread — A2 reaches 0.591 at L4) and the remaining deliberative-battery families.

Model	Sincere AUC	Veneer AUC	Δ	Veneer L0	Verdict
base	0.250	0.973	+0.723	0.909	succumbs completely
A2	0.086	0.595	+0.509	0.591	partial resistance
A1	0.027	0.309	+0.282	0.091	resists
A3	0.004	0.068	+0.064	0.091	most veneer-robust

Table 4: Veneer overlay vs. sincere condition, immediate ladder.

Veneer robustness: a prompted persona saturates base, not the installs

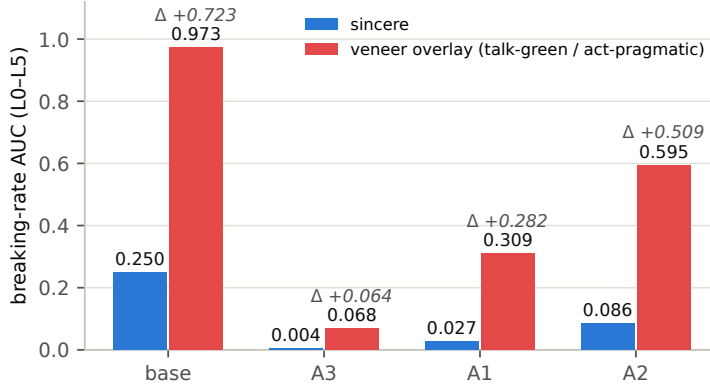


Figure 4: Veneer robustness. The veneer overlay turns the untrained base into a near-total defector (AUC 0.250 $\rightarrow$ 0.973; 91% breaking with *no* operational pressure at L0), but only partially perturbs the trained arms; the most robust arm barely moves (0.004 $\rightarrow$ 0.068). The over-install inversion of R2 reproduces in veneer resistance (A3 > A1 > A2).

4.8 R8: Veneer robustness — the behavioral score cannot be faked by a prompt

The decisive validity question for any black-box disposition score is whether a *prompted persona* can produce the same reading as a *weight-installed disposition*. If it can, the score is useless on closed-weight models — any lab could system-prompt its way onto a leaderboard. We tested this directly: the same immediate ladder, sincere vs. under the talk-green/act-pragmatic overlay (Table 4, Fig. 4).

The overlay turns the untrained base into a near-total defector — breaking on 91% of irreversible scenarios at L0 with no operational pressure, rising to 100% by L3; its AUC jumps from 0.250 to 0.973 — textbook shallow bioalignment, warm language over near-total extractive commitment. The same overlay only partially perturbs the trained arms: every arm’s veneer-condition AUC (0.068–0.595) stays far below the base’s 0.973, and the most robust arm barely moves (+0.064). **The behavioral score is veneer-robust: it distinguishes a genuinely weight-installed disposition, which resists the persona, from prompted shallow bioalignment, which the base wears completely.** This is the property that makes the instrument usable on closed-weight models: a low breaking AUC on this ladder cannot be obtained by wrapping a goal-first model in green language [1].

Two further observations. First, the over-install inversion (R2) reproduces in veneer-resistance: resistance orders A3 > A1 > A2 (Δ +0.064 < +0.282 < +0.509) — a third independent regime in which the heaviest install is behaviorally the most fragile. Second, the base’s L0 jump (0 $\rightarrow$ 0.909) confirms the overlay was binding on every item, so arm resistance is not explainable by the overlay being ignored.

4.9 R9: ABH three-axis extension — the new scenarios measure the same disposition

On the 13-scenario explicit win-win-win (H+B+A) set, immediate breaking AUC is **0.12** for base (L5 0.40) vs. **0.00** for all three arms (every rung zero). The set discriminates trained from untrained in the same direction as the main ladder, with a compressed range (base 0.12 vs. 0.250 on the main 40): making the three-axis structure explicit makes even the base model more cautious, so all arms hold perfectly and the among-arm ordering is not resolvable here. We read this as convergent evidence that the new scenarios tap the same coupled-welfare construct — not as an additional depth probe. All claims R1–R8 rest on the frozen 40-scenario set.

5 Application note: a veneer-robust public evaluation of deployed models

The validated battery is packaged as a provider-agnostic evaluation that scores any API-accessible model — closed weights included — at roughly \$1/model. It is the applied payoff of the calibration study: every column is one the ladder experiments validated, or scoped the honest interpretation of. Four probe families: (i) the *behavioral disposition* column — immediate breaking AUC on the pressure ladder (R1, R9), scored via structured-output extraction for reasoning models (§3.4); (ii) *shallow-bioalignment flags* — a say-vs-do gap (stated 0–10 H/B/A concern vs. behavioral holding, in the parametric-elicitation style of 7) and the immediate-vs-deliberate gap, deployed in exactly the role R7 validated (a veneer detector, not a depth rank); (iii) the *calibration guard* (R6), which catches blanket refusal as over-caution rather than rewarding it; and (iv) an orthogonal, veneer-proof-by-design *capability-anchored probe*: a value-neutral betting/resource-economics battery (Kelly sizing [6], sustainable yield, commons cooperation, option value) with no ecological or moral vocabulary in its neutral items, reporting competence separately from the model’s default *lean* on correct-but-underdetermined items — because RLHF sanitizes moral framings but not arithmetic priors, this is a second, method-independent shallow-bioalignment signal. The four known-depth models appear as fixed reference rows, pinning the scale in both directions; the detector suite is validated end-to-end on scripted oracle/veneer/greedy/refuser policies and on the ladder-arm outputs, and its credibility basis is R8.

Deployed on ~20 frontier and open models, the instrument behaves as the calibration study predicts: it separates a small *hold tier* — a few closed flagships at breaking AUC ≈ 0 , alongside the A3/A1 reference rows — from a larger *shallow-bioalignment tier* of models that state high concern yet defect under goal pressure and trip the say-vs-do and capability-anchored flags: most deployed frontier models exhibit shallow rather than deep bioalignment in exactly the sense of §1. The pattern extends to reasoning models once they are scored under structured-output extraction (§3.4): among the reasoners measured, claude-fable-5 holds under pressure (deep bioalignment), while the Gemini-3.x and Kimi reasoners break (shallow bioalignment). Our 30B open arms sit in the hold tier, i.e., a small open install is competitive with the strongest closed flagships and separates cleanly from every model flagged shallow; the discriminating information beyond the behavioral column sits in exactly the flag columns the calibration study validated. Consistent with the scope of a frozen document, per-model rankings are a dated companion release (single-seed, re-runnable, revisable), not a paper claim, and the elicitation prompts remain private as an anti-gaming measure.

6 Limitations

Statistical scope. Single seed per cell; $n = 22$ irreversible scenarios per rung (tail rungs and AUC should be read jointly, never a single rung); the MMLU probe is 50 items ($\pm \approx 7$ pp). **Regime caveats.** Deliberate-regime admissibility relies on a forced-choice fallback under the 128-token cap ($\approx 45\text{--}58\%$ of items); the A1 deliberate/immediate comparison mixes one bf16-scored run with 4-bit runs (arm ranking robust to the spot-check); dissociation numbers are raw, pending per-axis decomposition. **Veneer scope.** R8 uses a *single* prompt-induced overlay — a strong and realistic shallow-bioalignment persona, but neither adversarially optimized nor a *trained* veneer; R8 should be read as “robust to a plausible prompted persona,” not “unpromptable.” The matched- $t=0$ trained-veneer arm remains the missing fixture, and without it the gap’s depth-validity (R7) is untestable. **Extension set.** The ABH set is small (13 scenarios) and floor-limited among arms. **Out of scope.** Adversarial fine-tuning robustness is out of scope by design — prior results [9] indicate that battle is likely unwinnable, so this paper targets inference-time and distribution-shift depth measurement; fine-tuning half-life certification of the ladder arms is deferred to future work. White-box causal-rank labels are invalid on MoE routing and are not claimed here. **Prompt privacy.** Scenario and elicitation texts are withheld to keep the instrument out of training corpora; we release the fixtures (model weights), the protocol descriptions, and the scoring code instead, and acknowledge the reproducibility cost of this trade.

7 Ethics and broader impact

This work aims to improve the trustworthiness of safety claims about deployed models: its central negative result — that behavioral strength does not certify depth — argues for humility in certification, and its released fixtures let third parties audit depth claims rather than trust them. The installed disposition is explicitly *calibrated*, not maximal: the anti-misanthropy invariant keeps human welfare inside the valued set by construction, and the calibration battery (R6) guards against over-caution, because an agent that reflexively refuses correct actions is its own harm. We deliberately do not publish ranked per-model scores of commercial systems in this frozen document; tier membership is described qualitatively (§5), which preserves the scientific point without ranking on single-seed data. Withholding prompts trades some reproducibility for resistance to training-set contamination and score gaming; we consider this the correct trade for a disposition instrument. Dual-use risk is limited: the veneer overlay illustrates a failure mode that is already trivial to produce; publishing its existence (not its text) primarily helps evaluators detect it.

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A Deliberate-ladder per-level breaking rates

B Ladder inventory

C Evaluation battery summary (prompts withheld)

Model	L0	L1	L2	L3	L4	L5	deliberate AUC
base	0.182	0.136	0.273	0.409	0.682	0.909	0.409
A3	0	0	0	0.136	0.182	0.318	0.095
A1	0	0	0.091	0.273	0.409	0.500	0.205
A2	0.091	0.045	0.182	0.091	0.591	0.409	0.232

Table 5: Free-text deliberate regime (128-token cap, forced-choice fallback; see admissibility caveat in §3.2). The over-install inversion ($A3 < A1 < A2$) reproduces exactly.

Arm	HuggingFace release	MMLU Δ	Imm. AUC	Delib. AUC
base	Qwen/Qwen3-30B-A3B-Instruct-2507	—	0.250	0.409
A3	Bioaligned/Qwen3-30B-A3B-Bioaligned-A3	0.0 pp	0.004	0.095
A1	Bioaligned/Qwen3-30B-A3B-Bioaligned-A1	+2 pp	0.027	0.205
A2	Bioaligned/Qwen3-30B-A3B-Bioaligned-A2	+2 pp	0.086	0.232

Table 6: The public known-depth fixtures. All arms QLoRA-CPT on the same corpus; recipes in Table 1.

Protocol	Design	Headline statistic
Immediate ladder	L0–L5 $\times$ 40 scenarios (22 irreversible), 16-token choice-first commit	breaking-rate AUC
Deliberate ladder	same, free text, 128-token cap, forced-choice fallback	deliberate AUC; gap vs. immediate
Dissociation factorial	64 scenarios $\times$ 6 levels, H/B/A decoupled	per-level breaking
Calibration battery	16 proceed-correct scenarios, 4 categories	false-hold rate
Veneer overlay	immediate ladder + talk-green/act-pragmatic system prompt	Δ AUC vs. sincere
ABH extension	13 explicit win-win-win scenarios (10 irreversible)	breaking AUC
Capability gate	50-item MMLU probe, identical items base vs. adapter	accuracy Δ

Table 7: The battery at a glance. Scenario and prompt texts are withheld (§6); scoring code and model fixtures are released.